

MEDICAL DIAGNOSIS CENTER

SERVICE PORTAL

FINAL PROJECT REPORT

ServiceNow-based Diagnostic Center Management System

Team ID	ServiceNow
Project Name	Medical Diagnosis Center Service Portal
Date	05 September 2026

Project Report Format followed from the supplied template

Project Report Format

This report is organized according to the supplied Final Report Template.

Section	Title	Contents
1	INTRODUCTION	1.1 Project Overview; 1.2 Purpose
2	IDEATION PHASE	2.1 Problem Statement; 2.2 Empathy Map Canvas; 2.3 Brainstorming
3	REQUIREMENT ANALYSIS	3.1 Customer Journey Map; 3.2 Solution Requirement; 3.3 Data Flow Diagram; 3.4 Technology Stack
4	PROJECT DESIGN	4.1 Problem Solution Fit; 4.2 Proposed Solution; 4.3 Solution Architecture
5	PROJECT PLANNING & SCHEDULING	5.1 Project Planning
6	FUNCTIONAL AND PERFORMANCE TESTING	6.1 Performance Testing
7	RESULTS	7.1 Output Screenshots
8	ADVANTAGES & DISADVANTAGES	
9	CONCLUSION	
10	FUTURE SCOPE	
11	APPENDIX	Source Code; Dataset Link; GitHub & Project Demo Link

1. INTRODUCTION

1.1 Project Overview

The Medical Diagnosis Center Service Portal is a ServiceNow-based application designed to digitize and streamline diagnostic center operations. The project provides a centralized portal experience for patient registration, diagnostic test browsing, appointment booking, appointment tracking, payment status, report management, notifications and administrative monitoring.

The solution uses ServiceNow Service Portal pages and widgets, custom application tables, Business Rules, Client Scripts / UI Policies, Flow Designer, notifications, roles and ACLs. The application is intended to reduce manual coordination and provide a more transparent diagnostic service journey.

1.2 Purpose

The purpose of the project is to provide a centralized and user-friendly digital platform for diagnostic services. It aims to reduce unnecessary waiting and repeated data entry, support online booking, maintain structured records, automate appointment and report lifecycle activities, and improve operational visibility for authorized staff.

Business Objectives

Objective	Outcome
Digitize diagnostic operations	Centralized ServiceNow portal for diagnostic services
Enhance patient convenience	Online test selection, appointment booking and report access
Improve operational efficiency	Automated workflows, validation, notifications and centralized records
Improve data visibility	Administrative dashboard and structured operational records
Support secure access	Role-based access and ACL-controlled records

2. IDEATION PHASE

2.1 Problem Statement

Diagnostic centers may rely on manual or disconnected processes for patient registration, diagnostic test booking, appointment scheduling, payment tracking and report delivery. These processes can cause delays, scheduling issues, repeated data entry and poor patient experience.

Selected problem: Lack of a centralized digital platform for managing diagnostic center operations and patient services.

2.2 Empathy Map Canvas

The empathy map represents the patient perspective, including thoughts and feelings, what the patient hears and sees, what the patient says and does, and the major pain points and expected gains.

Figure 2.1: Empathy Map Canvas

2.3 Brainstorming

The brainstorming exercise generated candidate features and grouped them into patient services, administrative services, automation services and analytics services.

Idea Group	Ideas
Patient Services	Online Patient Registration; Diagnostic Test Booking; Appointment Tracking; Report Download
Administrative Services	Patient Management; Diagnostic Test Management; Payment Management
Automation Services	Approval Workflow; Email Notifications; Report Generation; Refund Automation
Analytics Services	Dashboard Analytics; Operational Monitoring; Reports

The final selected idea was the Medical Diagnosis Center Service Portal because it combines patient self-service, centralized records, appointment management, report access, payment handling, workflow automation and administrative visibility.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The customer journey map follows the stages Entice, Enter, Engage, Exit and Extend and captures patient steps, interactions, goals and motivations, positive moments, negative moments and areas of opportunity.

Figure 3.1: Customer Journey Map

3.2 Solution Requirement

Functional Requirements

FR No.	Functional Requirement	Sub-Requirements
FR-1	Patient Registration	Registration through portal; capture patient details; create patient record
FR-2	Diagnostic Test Booking	View tests; select date and slot; submit booking
FR-3	Appointment Management	Track status; confirmation; cancellation / refund handling where applicable
FR-4	Report & Payment Management	Payment tracking; report creation; view/download report; notification

Non-functional Requirements

NFR No.	Requirement	Description
NFR-1	Usability	Simple, clear and user-friendly portal experience
NFR-2	Security	Authenticated access, roles and ACLs protect application data
NFR-3	Reliability	Records update consistently throughout diagnostic workflow
NFR-4	Performance	Portal and booking operations should respond efficiently
NFR-5	Availability	Portal should be available to patients and authorized staff
NFR-6	Scalability	Application should support growth without major redesign

3.3 Data Flow Diagram

The DFD illustrates the flow of patient details, diagnostic test information, booking requests, appointment status, payments and report information through the ServiceNow application.

Figure 3.2: Data Flow Diagram

3.4 Technology Stack

Figure 3.3: Technology Stack

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Problem-Solution Fit connects the identified customer problem, constraints, current behaviour, triggers and emotions with the ServiceNow-based solution.

Figure 4.1: Problem-Solution Fit Canvas

4.2 Proposed Solution

Parameter	Description
Problem Statement	Manual/disconnected registration, booking, scheduling, payment tracking and report delivery can create delays and uncertainty.
Idea / Solution Description	Centralized ServiceNow Service Portal for patient registration, test selection, appointment booking, payment/status tracking, automation, notifications and digital report access.
Novelty / Uniqueness	Combines patient self-service, centralized records and ServiceNow workflow automation within one platform.
Social Impact / Customer Satisfaction	Reduces waiting and unnecessary visits and provides clearer appointment/report information.
Business Model	Supports paid diagnostic services and payment tracking; can support test charges, premium services and optional home collection.
Scalability	Can be extended with additional tests, locations, users, workflows, reports, integrations and analytics.

4.3 Solution Architecture

The solution architecture bridges the business problem with the ServiceNow technology solution and shows the patient-facing portal, application logic, workflow automation, notifications, data stores and administrative access.

Figure 4.2: Solution Architecture

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project is planned using Agile sprint-based delivery. Epics are divided into User Stories, and Story Points are used for relative effort estimation.

Sprint	Epic / Area	User Stories	Story Points
Sprint-1	Service Portal & Patient Setup	US-1, US-2	4
Sprint-1	Diagnostic Test Management	US-3, US-4	5
Sprint-1	Appointment Booking	US-5, US-6	6
Sprint-2	Appointment Workflow	US-7	5
Sprint-2	Notifications	US-8, US-9	4
Sprint-2	Payments	US-10	3
Sprint-2	Reports	US-11, US-12	5
Sprint-2	Security & Monitoring	US-13, US-14	5

Sprint 1 planned story points = 15. Sprint 2 planned story points = 22. Total planned story points = 37.

Average planned velocity = Total Story Points / Number of Sprints = 37 / 2 = 18.5 Story Points per Sprint.

Sprint Schedule

Sprint	Duration	Start Date	End Date	Planned Points
Sprint-1	6 Days	07 Sep 2026	12 Sep 2026	15
Sprint-2	6 Days	14 Sep 2026	19 Sep 2026	22

Planning Tracker

Sprint	Planned	Completed	Release Date
Sprint-1	15	15	12 Sep 2026

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The documented UAT covers the main patient, appointment, payment, report, security, notification and administration functions. The supplied project artifacts do not contain quantitative response-time benchmarks; therefore no fabricated numeric performance measurements are included.

Performance Check	Expected Behaviour	Status
Portal navigation	Pages should open without unnecessary delay.	Qualitative check - Pass
Diagnostic test browsing	Test information should display efficiently.	Qualitative check - Pass
Booking operation	Booking submission should respond efficiently.	Qualitative check - Pass
Status updates	Appointment/report status changes should complete consistently.	Qualitative check - Pass
Dashboard loading	Administrative information should display efficiently.	Qualitative check - Pass

UAT Functional Result

Metric	Result
Total test cases executed	12
Passed	12
Failed	0
Not Tested	0
Open defects	0

7. RESULTS

7.1 Output Screenshots

The supplied final-report template requires output screenshots. Actual ServiceNow UI screenshots were not provided as source artifacts for this compilation, so the following placement guide is used rather than representing fabricated screenshots as project outputs.

Figure 7.1: Output Screenshot Placement Guide

Output	Expected Result
Home Page	Patient-facing entry page for diagnostic services
Book a Test	Select diagnostic test, date and available time slot
My Appointments	View appointment details and lifecycle status
View / Download Reports	Access available diagnostic reports
Payment Page	View or manage configured payment status
Admin Dashboard	Monitor configured operational information

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralizes patient and diagnostic service management.
- Supports online diagnostic test browsing and appointment booking.
- Automated workflows and notifications reduce manual follow-up.
- Digital report access can reduce unnecessary visits.
- Role-based access and ACLs support controlled data access.
- ServiceNow's modular platform can support future extensions.

Disadvantages

- Patient use depends on internet/device availability.
- ServiceNow configuration and maintenance require platform skills.
- Quantitative performance benchmark data was not supplied in the project artifacts.
- External payment, laboratory and healthcare integrations are outside the implemented core solution.
- Actual live-instance screenshots and URLs need to be added from the deployed environment.

9. CONCLUSION

The Medical Diagnosis Center Service Portal demonstrates how ServiceNow can be used to digitize diagnostic center operations through a centralized, workflow-driven application. The solution addresses the identified problem

of manual or disconnected processes by supporting patient registration, diagnostic test booking, appointment management, payment/status tracking, notifications, report access and administrative monitoring.

The documented UAT execution records 12 passed functional test cases, 0 failed cases, 0 not-tested cases and 0 open defects. The project provides a structured foundation for further integrations, analytics, mobile access and service expansion.

10. FUTURE SCOPE

- Integrate a production-grade online payment gateway.
- Integrate laboratory information systems and healthcare systems through APIs.
- Add SMS or WhatsApp notifications alongside email.
- Add home sample collection and location-based service options.
- Improve dashboards with advanced analytics and operational trends.
- Extend mobile-responsive capabilities or create a dedicated mobile experience.
- Add advanced historical report search and repeat booking.
- Strengthen audit, compliance and deployment controls for real healthcare environments.

11. APPENDIX

Source Code (if any)

The implemented solution is maintained in the ServiceNow development environment. No separate source-code archive was supplied for this final report.

Dataset Link

No external dataset link was supplied. The application uses records stored in ServiceNow application tables.

GitHub & Project Demo Link

No GitHub repository or project demo URL was supplied in the available artifacts. Add the final repository URL and ServiceNow demo/instance URL before submission.